



STM32F3 Technical Training

For reference only

Refer to the latest documents for details



I2S Peripheral



SPI / I2S mode switch

3

- The I2S protocol is used for audio data communication between a microcontroller/DSP and an audio Codec/DAC.
- I2S interface is implemented as a mode in the SPI peripheral.
- To switch from SPI to I2S mode:
 - Disable SPI peripheral (reset SPE bit in SPI_CR1 register)
 - Select I2S mode (set I2SMOD bit in SPI_I2SCFGR register)

I2S Features (1/2)

- Two I2Ss: Available on SPI2 and SPI3 peripherals.
- Two I2Ss extension added for Full-Duplex communication.
- Simplex/or Full duplex communication (transmitter and receiver)
- I2S2 and I2S3 operate in master or slave configuration.
- 8-bit programmable linear prescaler to support all standard audio sample frequencies from 8 kHz up to 192 kHz.
- Audio-frequency precision same as **high-density and XL-density devices.**
- Programmable data format (16-, 24- or 32-bit data formats).

I2S Features (1/2)

5

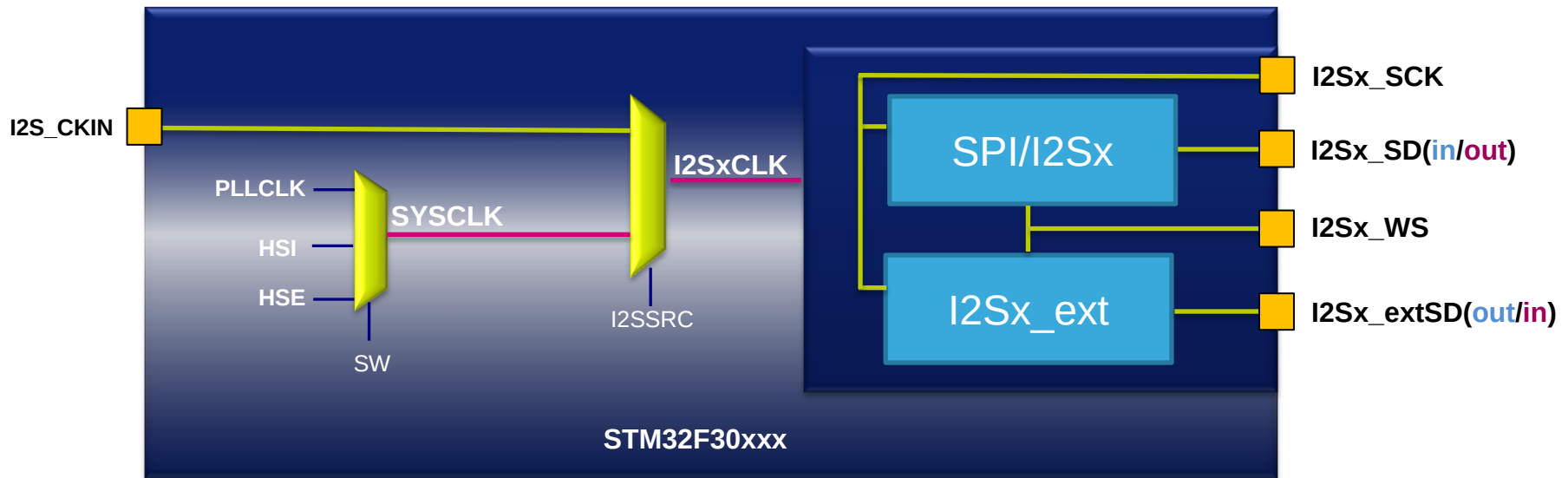
- Underrun flag in slave transmit mode, Overrun flag in receive mode and **new de-synchronization flag** in slave transmit/receive mode.
- Support for DMA: New DMA requests for I2S2_ext/I2S3_ext allows full duplex transfers.
- I2S protocols supported:
 - I2S Phillips standard.
 - MSB Justified standard (Left Justified).
 - LSB Justified standard (Right Justified).
 - PCM standard (with short and long frame synchronization on 16-bit channel frame or 16-bit data frame extended to 32-bit channel frame)
- **Master clock** may be output to drive an external audio component. Ratio is fixed at **256xFs** (where Fs is the audio sampling frequency).

 **The choice of the standard strongly depends on the external device and the audio data to be transmitted**

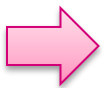
I2S full duplex block diagram(1/2)

6

- To support I2S full duplex mode, two extra I2S instances called extended I2Ss (I2S2_ext, I2S3_ext) are available in addition to I2S2 and I2S3. The first I2S full duplex interface is consequently based on I2S2 and I2S2_ext, and the second one on I2S3 and I2S3_ext.



Where x can be 2 or 3

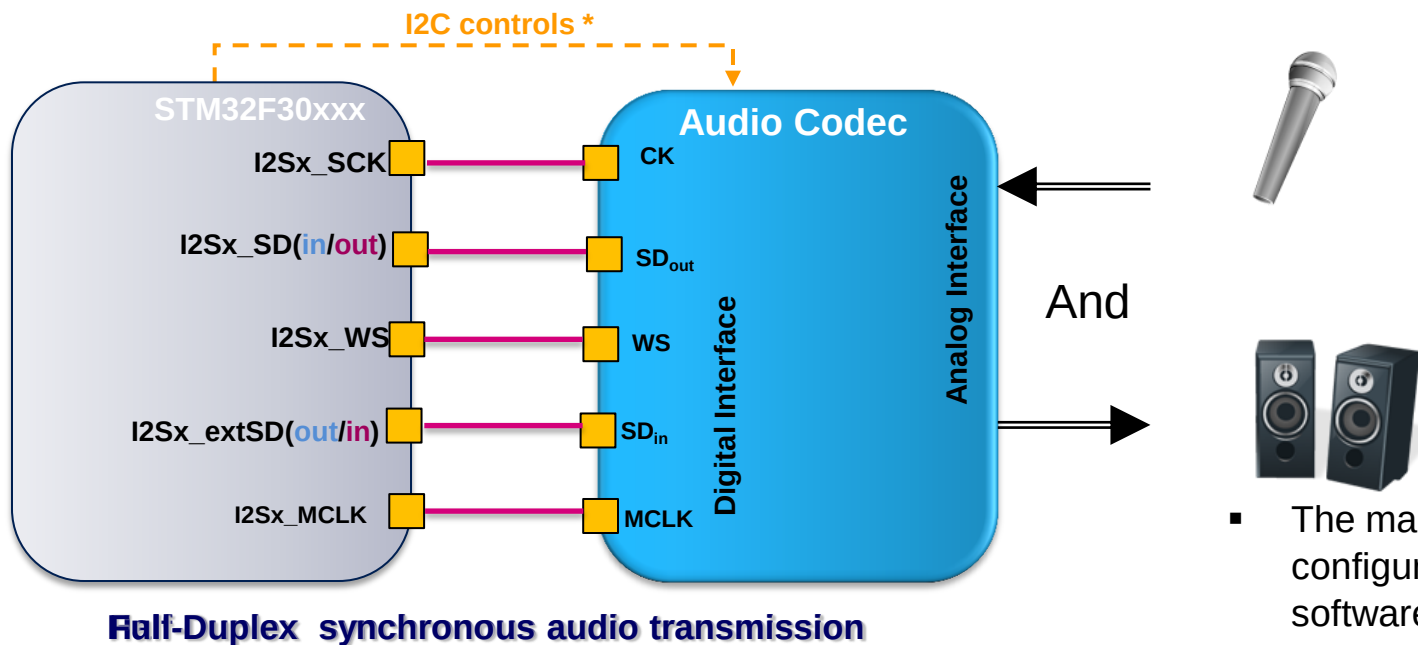


- I2Sx_ext can be used only in full duplex mode (**Always in slave mode**).
- Both I2Sx and I2Sx_ext can be configured as transmitters or receivers

Half/Full-Duplex Communication

7

- I2S configured in Half/Full-Duplex communication mode:

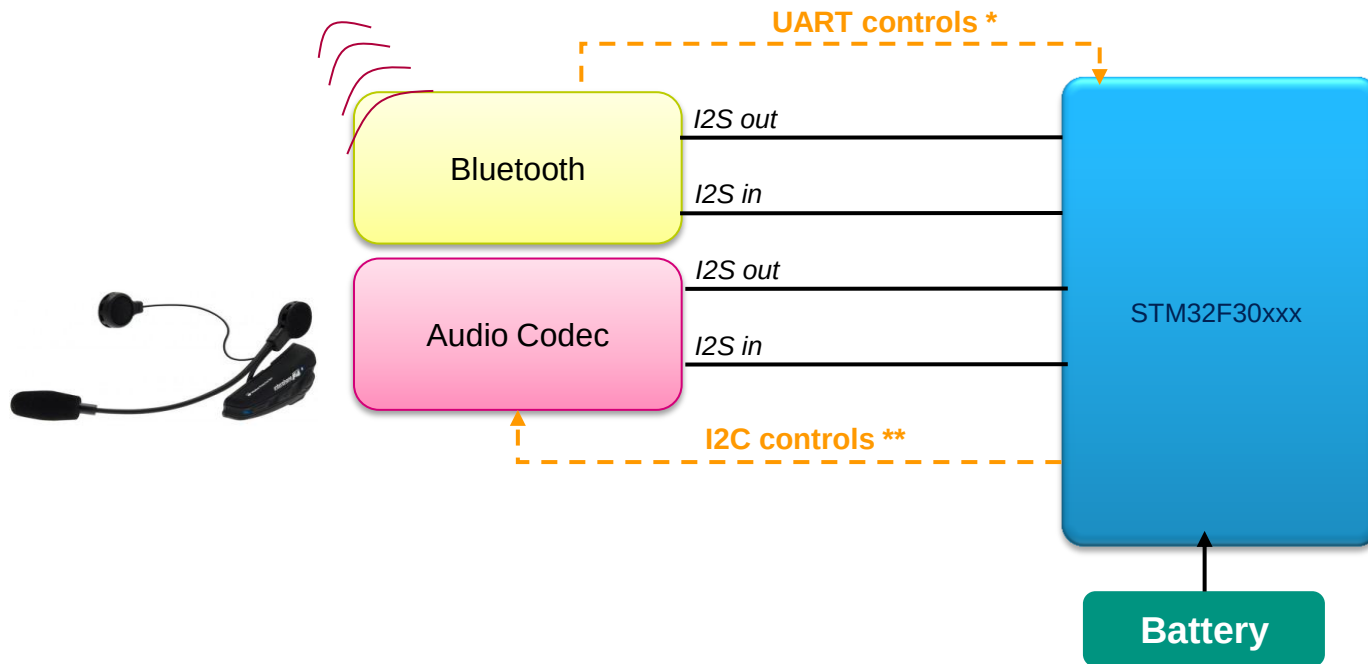


* Depends on the Codec control method

- The master and slave configuration is managed only by software. The master device is the CK and WS generator.
- The master/slave modes and transmit/receive directions can be switched dynamically by software.

2 x Full-Duplex Communication

8



Bluetooth Headset

* Depends on the Codec control method

** Depends on the Bluetooth control method

STM32F37xxx vs STM32F30xxx

9

Features	STM32F37xxx	STM32F30xxx
Instance	3 (I2S1, I2S2,I2S3)	2 (I2S2, I2S3)
Communication mode	Simplex	Simplex/full-duplex
External clock	No	Yes

- How many I2Ss are available in the STM32F30xxx and STM32F37xxx microcontroller?

- How to use I2Ss available in the STM32F30xxx in Full duplex mode?

- What are the standard audio frequencies supported by I2Ss?

- What are the different I2S error flags?



Thank you !

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